EP. 1

CODING INTERVIEW

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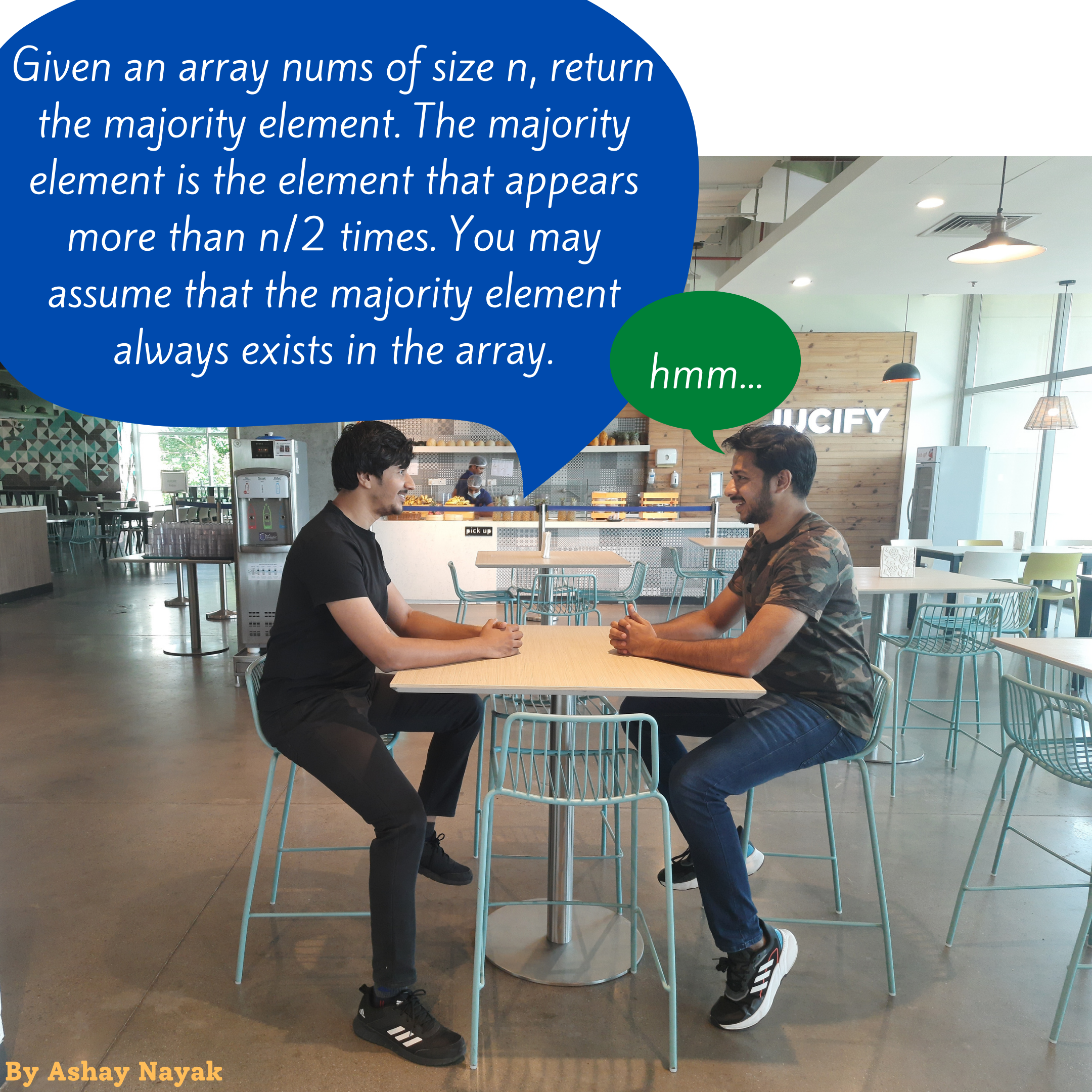

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EXAMPLES

Example 1:

Input: nums = [3,2,3]

Output: 3

Example 2:

Input: nums = [2,2,1,1,1,2,2]

Output: 2

APPROACH 1

Two For loops: Iterate through each element of the nums array. For each element, iterate again through the nums array and count the occurrence of that element. If count is greater than floor(n/2) then the element is the majority element.

TC → $O(n^2)$ and SC → $O(1)$

APPROACH 2

HashMap: Iterate through each element of the nums array and increment its count in the map. Here, key will be an element present in nums and value will be the count. Now, iterate through the map and return the element with a count value greater than floor(n/2).

TC → O(n) and SC → O(n)

APPROACH 3

Sorting: Sort the nums array and middle element is the answer.

But why?

As per the question, it is said to find the element that appears more than n/2 times, so after sorting, the majority element always takes the n/2th position.

For examples (0 based indexing):

A.) n=10, sorted nums = [1,2,3,3,4,4,4,4,4,4] → n/2 = 10/2 = element at 5th index is answer = 4

B.) n=11, sorted nums = [1,2,2,2,2,2,2,3,4,5, 6] → n/2 = 11/2 = element at 5th index is answer = 2

TC → O(nlogn) and SC → O(1)

APPROACH 4

Boyer-Moore Majority Voting Algorithm: Best solution

If you are looking to understand this algorithm with code for **interviews**

then check out my article link given in the comment below.

TC → O(n) and SC → O(1)

TO BE CONTINUED...

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